Artificial Intelligence

Course Objectives: The main objectives of this course are:

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To provide basic knowledge of Artificial Intelligence
To familiarize students with different search techniques
To acquaint students with the fields related to AI and the applications of AI

Introduction (4 hrs)
    Definition of Artificial Intelligence
    Importance of Artificial Intelligence
    AI and related fields
    Brief history of Artificial Intelligence
    Applications of Artificial Intelligence
    Definition and importance of Knowledge, and learning.

Problem solving (4 hrs)
    Defining problems as a state space search,
    Problem formulation
    Problem types, Well- defined problems, Constraint satisfaction problem,
    Game playing, Production systems.

Search techniques (5 hrs)
    Uninformed search techniques- depth first search, breadth first search, depth li
    Informed search techniques-hill climbing, best first search, greedy search, A* s

Knowledge representation, inference and reasoning (8 hrs)
    Formal logic-connectives, truth tables, syntax, semantics, tautology, validity,
    Propositional logic, predicate logic, FOPL, interpretation, quantification, horn
    Rules of inference, unification, resolution refutation system (RRS), answer extr
    Statistical Reasoning-Probability and Bayes' theorem and causal networks, reason

Structured knowledge representation (4 hrs)
    Representations and Mappings,
    Approaches to Knowledge Representation,
    Issues in Knowledge Representation,
    Semantic nets, frames,
    Conceptual dependencies and scripts

Machine learning (6 hrs)
    Concepts of learning,
    Learning by analogy, Inductive learning, Explanation based learning
    Neural networks,
    Genetic algorithm
    Fuzzy learning
    Boltzmann Machines

Applications of AI (14 hrs)
    Neural networks
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        Network structure
        Adaline network
        Perceptron
        Multilayer Perceptron, Back Propagation
        Hopfield network
        Kohonen network
    Expert System
        Architecture of an expert system
        Knowledge acquisition, induction
        Knowledge representation, Declarative knowledge, Procedural knowledge
        Development of expert systems
    Natural Language Processing and Machine Vision
        Levels of analysis: Phonetic, Syntactic, Semantic, Pragmatic
         Introduction to Machine Vision
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Practical: Laboratory exercises should be conducted in either LISP or PROLOG. Laboratory exercises must cover the fundamental search techniques, simple question answering, inference and reasoning.

References:


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E. Rich and Knight, Artificial Intelligence, McGraw Hill, 2009.
D. W. Patterson, Artificial Intelligence and Expert Systems, Prentice Hall, 2010.
P. H. Winston, Artificial Intelligence, Addison Wesley, 2008.
Stuart Russel and Peter Norvig, Artificial Intelligence A Modern Approach, Pearson,
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Evaluation Scheme: The question will cover all the chapters of the syllabus. The evaluation scheme will be as indicated in the table below: Chapter

Hour

Marks Distribution*

1

4

7

2

4

7

3

5

9

4

8

14

5

4

7

6

6

10

7

14

26

Total

45

80

*Note: There may be minor deviations in marks distribution